

Output_question-01:

```
Python 3.12.11 | packaged by conda-forge | (main, Jun 4 2025, 14:29:09) [MSC v.1943 64 bit (AMD64)]
Type "copyright", "credits" or "license" for more information.

IPython 9.15.0 -- An enhanced Interactive Python. Type '?' for help.

In [1]: %runfile C:/Users/AA/Documents/GitHub/python-exercises-Ayub-Abedini/session-05/question-01.py --wdir
Enter Password: 1234
Paeeword is invalid
Password must contain at least 8 characters

In [2]: %runfile C:/Users/AA/Documents/GitHub/python-exercises-Ayub-Abedini/session-05/question-01.py --wdir
Enter Password: ABCDabcd1234!@#$
Password is valid

In [3]: %runfile C:/Users/AA/Documents/GitHub/python-exercises-Ayub-Abedini/session-05/question-01.py --wdir
Enter Password: 1234ABC
Paeeword is invalid
Password must contain at least 8 characters

In [4]: %runfile C:/Users/AA/Documents/GitHub/python-exercises-Ayub-Abedini/session-05/question-01.py --wdir
Enter Password: 1234ABCDE
Pasword is invali
Password must contain a Lowercase character
Pasword is invali
Password must contain a spacial character ! @ # $

In [5]: %runfile C:/Users/AA/Documents/GitHub/python-exercises-Ayub-Abedini/session-05/question-01.py --wdir
Enter Password: asdfghjkl
Pasword is invali
Password must contain a Uppercase character
Pasword is invali
Password must contain a Number
Pasword is invali
Password must contain a spacial character ! @ # $
```

Output_question-02:

```
Python 3.12.11 | packaged by conda-forge | (main, Jun 4 2025, 14:29:09) [MSC v.1943 64 bit (AMD64)]
Type "copyright", "credits" or "license" for more information.

IPython 9.15.0 -- An enhanced Interactive Python. Type '?' for help.

In [1]: %runfile C:/Users/AA/Documents/GitHub/python-exercises-Ayub-Abedini/session-05/question-02.py --wdir
Enter a string: programming
progamin

In [2]: %runfile C:/Users/AA/Documents/GitHub/python-exercises-Ayub-Abedini/session-05/question-02.py --wdir
Enter a string: Ayub Abedini
Ayub edin

In [3]: %runfile C:/Users/AA/Documents/GitHub/python-exercises-Ayub-Abedini/session-05/question-02.py --wdir
Enter a string: 11111222223333aaaa333ddd
123ad
```

Output_question-03:

```
Python 3.12.11 | packaged by conda-forge | (main, Jun 4 2025, 14:29:09) [MSC v.1943 64 bit (AMD64)]
Type "copyright", "credits" or "license" for more information.

IPython 9.15.0 -- An enhanced Interactive Python. Type '?' for help.

In [1]: Enter a string: Python 123! ABC
Letters: 9
Uppercase: 4
Lowercase: 5
Digits: 3
Spaces: 2
Special characters: 1

In [2]: %runfile C:/Users/AA/Documents/GitHub/python-exercises-Ayub-Abedini/session-05/question-03.py --wdir
Enter a string: Ayub Abedini
Letters: 11
Uppercase: 2
Lowercase: 9
Digits: 0
Spaces: 1
Special characters: 0

In [3]: %runfile C:/Users/AA/Documents/GitHub/python-exercises-Ayub-Abedini/session-05/question-03.py --wdir
Enter a string: AYUB abedini 123 !@#
Letters: 11
Uppercase: 4
Lowercase: 7
Digits: 3
Spaces: 3
Special characters: 3
```

Output_question-04:

```
Python 3.12.11 | packaged by conda-forge | (main, Jun 4 2025, 14:29:09) [MSC v.1943 64 bit (AMD64)]
Type "copyright", "credits" or "license" for more information.

IPython 9.15.0 -- An enhanced Interactive Python. Type '?' for help.

In [1]: %runfile C:/Users/AA/Documents/GitHub/python-exercises-Ayub-Abedini/session-05/question-04.py --wdir
Enter a your sentence: python is easy and python is powerful and python is popular
python -> 3

In [2]: %runfile C:/Users/AA/Documents/GitHub/python-exercises-Ayub-Abedini/session-05/question-04.py --wdir
Enter a your sentence: Ayub Ali Ali Reza Ali Ahmad
Ali -> 3
```

Output_question-05:

```
Python 3.12.11 | packaged by conda-forge | (main, Jun 4 2025, 14:29:09) [MSC v.1943 64 bit (AMD64)]
Type "copyright", "credits" or "license" for more information.

IPython 9.15.0 -- An enhanced Interactive Python. Type '?' for help.

In [1]: %runfile C:/Users/AA/Documents/GitHub/python-exercises-Ayub-Abedini/session-05/question-05.py --wdir
Enter your sentence: Python programming is extremely interesting
Longest word: programming
Length: 11

In [2]: %runfile C:/Users/AA/Documents/GitHub/python-exercises-Ayub-Abedini/session-05/question-05.py --wdir
Enter your sentence: I am learning Python programming every day
Longest word: programming
Length: 11
```

Output_question-06:

```
Python 3.12.11 | packaged by conda-forge | (main, Jun 4 2025, 14:29:09) [MSC v.1943 64 bit (AMD64)]
Type "copyright", "credits" or "license" for more information.

IPython 9.15.0 -- An enhanced Interactive Python. Type '?' for help.

In [1]: %runfile C:/Users/AA/Documents/GitHub/python-exercises-Ayub-Abedini/session-05/question-06.py --wdir
Enter your sentence: This is password attack and another password attack
password -> 2
attack -> 2

In [2]: %runfile C:/Users/AA/Documents/GitHub/python-exercises-Ayub-Abedini/session-05/question-06.py --wdir
Enter your sentence: I saw a hack and a scam. This hack was dangerous. The password was also stolen.
hack -> 2
password -> 1
```

Output_question-07:

```
Python 3.12.11 | packaged by conda-forge | (main, Jun 4 2025, 14:29:09) [MSC v.1943 64 bit (AMD64)]
Type "copyright", "credits" or "license" for more information.

IPython 9.15.0 -- An enhanced Interactive Python. Type '?' for help.

In [1]: %runfile C:/Users/AA/Documents/GitHub/python-exercises-Ayub-Abedini/session-05/question-07.py --wdir
Enter a your text: aaabbcccdad
a3b2c4d1a2

In [2]: %runfile C:/Users/AA/Documents/GitHub/python-exercises-Ayub-Abedini/session-05/question-07.py --wdir
Enter a your text: a3b2c4d1a2
a131b121c141d111a1a1
```

Output_question-08:

```
Python 3.12.11 | packaged by conda-forge | (main, Jun 4 2025, 14:29:09) [MSC v.1943 64 bit (AMD64)]
Type "copyright", "credits" or "license" for more information.

IPython 9.15.0 -- An enhanced Interactive Python. Type '?' for help.

In [1]: %runfile C:/Users/AA/Documents/GitHub/python-exercises-Ayub-Abedini/session-05/question-08.py --wdir
Enter a your text: Python is a powerful programming language and Python is easy to learn
Total characters: 69
Total words: 12
Total letters: 58
Total digite: 0
Total uppercases: 2
Total lowercases: 56
Longes word: programming
Shortest word: a
Most repeated character:
Most repeated word: Python

In [2]: %runfile C:/Users/AA/Documents/GitHub/python-exercises-Ayub-Abedini/session-05/question-08.py --wdir
Enter a your text: I am learning Python and I practice Python every day
Total characters: 52
Total words: 10
Total letters: 43
Total digite: 0
Total uppercases: 4
Total lowercases: 39
Longes word: learning
Shortest word: I
Most repeated character:
Most repeated word: I
```

Output_question-09:

```
Python 3.12.11 | packaged by conda-forge | (main, Jun 4 2025, 14:29:09) [MSC v.1943 64 bit (AMD64)]
Type "copyright", "credits" or "license" for more information.

IPython 9.15.0 -- An enhanced Interactive Python. Type '?' for help.

In [1]: %runfile C:/Users/AA/Documents/GitHub/python-exercises-Ayub-Abedini/session-05/question-09.py --wdir
Enter your username: Ayub Abedini
Enter your password: 5555555555
Wrong username or password
Attempts remaining: 2
Enter your username: Ayub Abedini
Enter your password: 5555555555
Wrong username or password
Attempts remaining: 1
Enter your username: Ayub Abedini
Enter your password: 123456789
Login successful
```

Output_question-10:

```
Python 3.12.11 | packaged by conda-forge | (main, Jun 4 2025, 14:29:09) [MSC v.1943 64 bit (AMD64)]
Type "copyright", "credits" or "license" for more information.

IPython 9.15.0 -- An enhanced Interactive Python. Type '?' for help.

In [1]: %runfile C:/Users/AA/Documents/GitHub/python-exercises-Ayub-Abedini/session-05/question-10.py --wdir
Enter sentence 1: I love Python programming
Enter sentence 2: Python is a powerful programming language
Common words: Python
Common words: programming

In [2]: %runfile C:/Users/AA/Documents/GitHub/python-exercises-Ayub-Abedini/session-05/question-10.py --wdir
Enter sentence 1: Python Python programming
Enter sentence 2: Python is good
Common words: Python
Common words: Python
```